

Deriving the model for the mean of the stochastic process
interval Δt

$$Pr(\text{birth}) = b\Delta t$$

$$Pr(\text{death}) = m\Delta t$$

$N(t)$ individuals

How many births in Δt ? What distribution?

$$B \sim \text{Binomial}(N, b\Delta t)$$

$$D \sim \text{Binomial}(N, m\Delta t)$$

} scaling up step

$$N(t+\Delta t) = N(t) + B - D$$

$$E[B] = Nb\Delta t \quad E[D] = Nm\Delta t$$

$$\begin{aligned} E[N(t+\Delta t)] &= N(t) + E[B] - E[D] \\ &= N(t) + Nb\Delta t - Nm\Delta t \end{aligned}$$

$$E[\Delta N] = E[N(t+\Delta t) - N(t)] = (b-m)N\Delta t$$

$$\lim_{\Delta t \rightarrow 0} \frac{E[\Delta N]}{\Delta t} = (b-m)N(t)$$

$$\boxed{\frac{dE[N(t)]}{dt} = (b-m)N(t). \quad \text{deterministic model,}}$$

What about variance? ... in Δt

Variance of a binomial

$$\text{Var}(x) = np(1-p)$$

In Δt at $N(t)$,

$$\text{Var}(B) = N(t)b\Delta t(1-b\Delta t) = N(t)b\Delta t - N(t)b^2\Delta t^2$$

$$\approx N(t)b\Delta t \quad \text{as } \Delta t \rightarrow 0$$

$$\text{Var}(D) \approx N(t)m\Delta t$$

$$\text{Var}(\Delta N) = \text{Var}(B) + \text{Var}(D)$$

$$\Delta N = N(t) + B - D$$

$$= N(t)b\Delta t + N(t)m\Delta t$$

$$= (b+m)N(t)\Delta t$$

Stochastic approximation model

Mean change + variance in change

Deterministic skeleton + ^{noise}
(stoch component)

$$N(t+\Delta t) = N(t) + (b-m)N(t)\Delta t + \sqrt{(b+m)N(t)\Delta t} Z$$

$$Z \sim \text{Normal}(0,1)$$